Mid-term presentation — FINAL delivery script

Parv: slides 1–6 — 893 words, 5:00 (unchanged; this is the version you timed)
Mayank: slides 7–13 — 600 words, 4:40

Budgets are set from your *measured* speaking rates, not a generic one: Parv runs at about 180 words a minute, Mayank at about 130. That difference is the whole reason the halves came out uneven — not effort. So Mayank's section is two-thirds the word count of Parv's, and it should land in the same five minutes.

The slides are unchanged.

Timings are spoken content only. `[...]` are stage directions. Lines marked ✂ are safe to drop live if you are behind.

PARV — slides 1 to 6 — target 5:00

Slide 1 — Title 0:40

Good morning. Our project is on motor subtype progression in Parkinson's disease, using the PPMI dataset.

There's a standard clinical rule that labels a patient as tremor-dominant or PIGD at a given visit. That rule works on one visit at a time, and it works well for that. What we wanted to understand is what happens *across* visits — because when a patient's recorded subtype changes, that could be the disease progressing, or it could just be the examination coming out differently that day.

Separating those two things is the whole project.

Slide 2 — One number, cut into three 0:50

The subtype comes from a single number. Eleven tremor items, five gait and postural items, take the two averages and divide. Above 1.15 you're tremor-dominant, below 0.90 you're PIGD, and in between you're Indeterminate.

[*point at the bar*]

The thing to notice is this picture. That middle band is 0.25 units wide — but the ratio itself runs from zero to about eleven. So the band is very narrow relative to the scale it sits on. Please hold onto that, because it comes back twice.

And the rule is a snapshot. It tells you where a patient is today. It has nothing to say about whether a change between two visits is real.

Slide 3 — The dataset we built 0:45

We didn't take the subtype column off the shelf — we rebuilt it from the raw item scores, so that we controlled every decision. Sporadic Parkinson's only, and only OFF-medication examinations, because that's what the standard definition uses.

That gives about a thousand patients and four and a half thousand labelled visits. For anything longitudinal you need at least two labelled visits, and that's 934 patients. That number is the cohort for everything that follows.

✂ We validated our labels against PPMI's own — we agree on every single visit for the two-class column, and on 99.87% for an archived three-class one.

And the box on the right. "Unable to Rate" is recorded as the number 101, sitting in the same column as a real score of 0 to 4. If you average that in, one entry pushes the PIGD score to about twenty, and the patient gets labelled PIGD when they're actually tremor-dominant. There are over three thousand of those on the postural stability item alone. Finding that early is a large part of why we trust the rest.

Slide 4 — What the records actually show 0:50

This is just counting — no model yet. Take every pair of consecutive labelled visits and ask where the patient went.

Two things jump out. The middle row: if a patient is Indeterminate today, there's only a 19% chance they're Indeterminate at the next visit. They're much more likely to be somewhere else. So Indeterminate isn't really behaving like a category — it looks more like a region patients pass through.

And PIGD isn't a one-way door either. Almost 30% of PIGD visits are followed by something that isn't PIGD, and 18% go straight back to tremor-dominant. People often think of PIGD as the endpoint; the record doesn't support that.

But — and this matters — there is still real drift. Changes towards PIGD outnumber the other direction 253 to 141, and that's highly significant. So something genuine is going on. It's just buried in a lot of movement.

Slide 5 — A single recorded change is weak evidence 1:10

⚑ **Key slide — do not rush this one**

This slide is the heart of the motivation, so I'll spend a moment on it.

A third of all the subtype changes we observe reverse immediately — at the very next visit the patient is back where they were. That is not what progression looks like.

Second number. If we take apparent conversions to PIGD and add one requirement — the new subtype has to still be there at the next visit — half of them disappear. 263 becomes 127.

And the reason is the third number. Within a single patient, the ratio bounces around with a standard deviation of 1.22, and the band we're asking it to sit inside is 0.25 wide. It's about five times too noisy for the question we're asking it. In 73% of consecutive visits, a patient's own ratio moves further than the entire width of the middle band.

So if you happen to sit near a threshold, you will cross it now and then for reasons that have nothing to do with your disease.

That's the problem we needed a model for.

Slide 6 — The simple model failed 1:00

We didn't jump straight to a complicated model. We started with the simplest thing: a Markov chain on the observed label. That model says the current label is all you need — where the patient came from shouldn't matter.

So we tested exactly that. Take every visit where the patient is currently labelled TD, and split them by where they came from. If they arrived at TD from TD, there's an 8.6% chance they're PIGD next visit. If they arrived at TD from PIGD, it's 39.3%.

Same label today. Four and a half times the risk tomorrow.

Chi-squared is 156, and it's not just TD — the test rejects for all three labels.

So the observed label is not a complete description of the patient. Something is being remembered that the label doesn't capture. And that is precisely the pattern you get when there's a real underlying state and you're only seeing a noisy reading of it.

That's what took us to the hidden Markov model — not because it was next in a textbook, but because the data told us the simpler one was wrong.

[*handover*] I'll hand over to Mayank.

⏱ **Checkpoint: hand over at 5:00.**

MAYANK — slides 7 to 13 — target 4:40

> **Read this before you practise.** You speak at about 130 words a minute — that is a good, clear pace and you should not try to speed up. Instead this script is written to *your* rate: it is 600 words, which lands at 4:40. The previous one was 900, which is why it came out at seven minutes.
>
> Three rules: 1. **Never read a table aloud.** Say what it means; they can read. 2. **One sentence, one idea.** Full stop, breathe, next. 3. **If you fumble a number, carry on.** Don't restart the sentence — the restarts are where the two minutes went.

Slide 7 — Separating the state from the label 0:45

Thanks. Here's the idea.

There's an underlying motor state — the top row. It moves gradually, and we never see it.

What we see is the bottom row. That's what the examination wrote down.

[*point at the third column*]

Third column. The state is Indeterminate. The exam recorded PIGD.

So the model estimates two things at once. How fast the real state changes. And how reliably it gets recorded. The earlier models couldn't express that second part.

One caveat. We impose a minimum recording reliability, or the fit degenerates. That's our choice, not the data's.

Slide 8 — What the model estimates 0:50

Three results — and these are the model's estimates, not measurements.

First, it recovered an ordering we never gave it. Patients move *through* Indeterminate. Going straight from TD to PIGD is sixteen times rarer. PIGD back to TD comes out at zero.

Second, reliability. TD is recorded right 94% of the time. PIGD, 84%. Indeterminate, 50% — and to be precise, **50% is the floor we imposed**, so it's an upper bound. Unconstrained, the data pushes it to 0.36. Essentially uninformative.

Third, the key one. The labels suggest a TD patient stays TD about three and a half years. The model says the underlying state lasts around ten.

So most of the churn is measurement, not the patient.

Slide 9 — Validation 0:55

⚑ **Key slide — the examiners will focus here**

This is the strictest validation we can do.

We hold out one patient completely. Refit on the other 933. Then predict their visits one at a time — and to predict a visit, the model only sees what came before.

That's 3,361 predictions, at **75.4%**.

Now the honest reading. The HMM wins, but by under a point over a plain Markov chain, and about three over just saying nothing changes. The real separation is the Brier score, which judges the whole distribution rather than the top choice.

And the caveat that matters most. We predict the next label *our own rule* produces. That's a noisy record. So this is agreement with a record — not accuracy about disease progression, and not clinical validation.

Slide 10 — Two stress tests 0:45

Then we tried to break that number. Two ways.

First, we narrowed the Indeterminate band. Accuracy rose, to 78%. But the persistence baseline rose more — so our advantage over "nothing changes" halved. We'd only deleted the hardest class. The number improved; the model didn't.

Second, five latent states instead of three. No better. Only 404 of 934 folds converged, and six starting points gave six different answers. That model isn't identified by this data.

Two negative results, one lesson. Accuracy alone is a misleading way to compare these models. That's why we quote the gap to the baseline.

⏱ **Checkpoint: leave this slide at about 8:20.**

Slide 11 — The continuous ratio 0:30

We also built the proposal's Bayesian growth model, on the continuous ratio.

It gives one thing the HMM can't — a progression rate in clinical units. About 0.09 a year.

It loses on prediction, for the same reason as everything else: its spread is about five times the width of the band.

So we keep both. And the box at the bottom is our overall reading — the noise and the signal are both real, at comparable size.

Slide 12 — Where this goes next 0:40

Four directions.

First, the ratio throws away severity. Two over four and ten over twenty are both 0.5 — very different patients. So we want to model tremor and gait separately. That also recovers 725 visits where the ratio doesn't exist at all.

Second, every model here assumes a constant rate. We want to compare early and late follow-up. That's a hypothesis, not a result.

Third, whether patients fall into distinct progression patterns at all.

Fourth, understanding *when* the model fails. The five-state result is why that matters.

Slide 13 — Close 0:15

To close. Subtype progression is a gradual underlying process, seen through an imperfect measurement.

So a single recorded change is weak evidence. A sustained one is much stronger.

Thank you — happy to take questions.

Timing sheet

Slide	Who	Target	Cumulative
1 Title	Parv	0:40	0:40
2 One number	Parv	0:50	1:30
3 Dataset	Parv	0:45	2:15
4 Records	Parv	0:50	3:05
5 Weak evidence	Parv	1:10	4:15
6 Markov failed	Parv	1:00	**5:15 → handover**
7 State vs label	Mayank	0:45	6:00
8 What it estimates	Mayank	0:50	6:50
9 Validation	Mayank	0:55	7:45
10 Stress tests	Mayank	0:45	**8:30**
11 Continuous ratio	Mayank	0:35	9:05
12 Future work	Mayank	0:45	9:50
13 Close	Mayank	0:15	**10:05**

Two checkpoints to watch on a phone timer:

- Handover at **5:00–5:15**. If Parv is past 5:30, he drops the ✂ line on slide 3.
- Leaving slide 10 at **8:15–8:30**. If Mayank is past 8:45, he drops the ✂ line on

slide 11 and delivers slide 12 as four short headlines only — one sentence each, no elaboration.

If Mayank is still over after practising

Do these in order.

1. **Slide 11 becomes three sentences.** "We also built the proposal's growth model. It gives a progression rate in clinical units — about 0.09 a year — which the HMM can't. It loses on prediction, so we keep both for different reasons." Saves ~20 s.
2. **Slide 12 becomes headlines.** Say only the bolded phrase from each of the four rows, plus one clause. Saves ~20 s.

3. **Move slide 7 to Parv.** It follows directly from his slide 6 — "that's what took us to the hidden Markov model, and here's how it works" — so it's a natural join, and Parv has proven he holds time. This moves 45 s across and leaves Mayank six slides instead of seven.

Option 3 is the biggest single lever. Only use it if 1 and 2 aren't enough, because it means re-rehearsing the handover.

Three numbers Mayank must not fumble

- **934 patients / 3,361 predictions / 75.4%**
- **+3.1** → **+1.2** — the margin over baseline collapsing when the band narrows
- **404 of 934** folds converged with five latent states

Two language traps

- Slide 8: say **"underlying state"** or **"latent state"** — never "true disease state".
- Slide 9: say **"agreement with the next recorded label"** — never "accuracy at predicting progression".

If you're asked something you don't know, say so and say how you'd test it. "We haven't tested that, but here's how we would" is a perfectly good answer at a mid-term.